Results for Eedi

Three training-graph variants; six models per variant; N = 5, 10, ..., 50.

Higher is better for Ada, NOV and Ep. Bold = best; underline = second best within each row.

Rankings use unrounded values. All 18 model runs completed for this dataset.

Training-graph variant	Retained exfr edges	Share of source	Mean forgetting
1. Remove exfr0.00 training edges	532,598 / 664,548	80.1%	0.369
2. Weighted nonzero sampling, up to 400/student	277,020 / 664,548	41.7%	0.504
3. Weighted nonzero sampling, up to 200/student	140,118 / 664,548	21.1%	0.544

Main model across the three variants

Variant	Ada@10	Ada@50	NOV@50	Ep0.5@50	Ep0.7@50
Remove exfr0.00 training edges	0.677	0.665	0.982	0.004	-0.730
Weighted nonzero sampling, up to 400/student	0.673	0.662	0.983	-0.142	-0.738
Weighted nonzero sampling, up to 200/student	0.676	0.664	0.982	0.014	-0.613

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 532,598 / 664,548 (80.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.685	0.656	**0.693**	0.671	0.677	0.686
10	0.677	0.656	**0.685**	0.666	0.673	0.675
15	0.673	0.656	**0.676**	0.663	0.671	0.673
20	0.671	0.655	**0.672**	0.662	0.669	0.671
25	0.670	0.656	**0.671**	0.662	0.668	0.670
30	0.668	0.656	**0.670**	0.661	0.667	0.669
35	0.666	0.655	**0.669**	0.661	0.666	0.668
40	0.666	0.655	**0.667**	0.661	0.665	0.666
45	0.665	0.655	**0.667**	0.660	0.664	0.666
50	0.665	0.655	**0.666**	0.660	0.664	0.666

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.986	0.985	0.987	**0.993**	0.987	0.985
10	0.984	0.986	0.983	**0.990**	0.985	0.983
15	0.982	0.984	0.985	**0.989**	0.985	0.981
20	0.982	0.984	0.984	**0.988**	0.983	0.982
25	0.982	0.984	0.983	**0.987**	0.983	0.981
30	0.983	0.983	0.984	**0.987**	0.983	0.982
35	0.983	0.983	0.983	**0.986**	0.982	0.982
40	0.982	0.983	0.983	**0.985**	0.982	0.982
45	0.982	0.984	0.983	**0.985**	0.983	0.981
50	0.982	0.984	0.984	**0.985**	0.983	0.981

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 532,598 / 664,548 (80.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.096	0.095	**0.186**	0.063	0.117	-0.036
10	0.146	0.194	**0.316**	0.148	0.252	-0.010
15	0.172	0.164	**0.671**	0.172	0.272	0.103
20	0.301	0.122	**0.706**	0.196	0.268	0.117
25	0.313	0.084	**0.635**	0.216	0.206	0.279
30	0.254	0.125	**0.474**	0.223	0.252	0.174
35	0.308	0.099	**0.506**	0.223	0.256	0.157
40	0.313	0.136	**0.406**	0.229	0.268	0.096
45	**0.371**	0.139	0.266	0.231	0.313	-0.077
50	0.336	0.085	**0.424**	0.228	0.323	0.061

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.014	0.011	**0.140**	-0.004	0.058	-0.102
10	0.018	0.036	**0.198**	0.019	0.113	-0.115
15	-0.025	-0.061	**0.546**	-0.007	0.082	-0.076
20	0.064	-0.142	**0.596**	-0.025	0.046	-0.093
25	0.039	-0.211	**0.540**	-0.035	-0.037	0.037
30	-0.032	-0.235	**0.411**	-0.053	-0.021	-0.119
35	0.006	-0.303	**0.422**	-0.070	-0.037	-0.145
40	-0.001	-0.313	**0.253**	-0.076	-0.048	-0.155
45	0.043	-0.334	**0.109**	-0.085	-0.044	-0.421
50	0.004	-0.385	**0.199**	-0.097	-0.063	-0.552

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 532,598 / 664,548 (80.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.051	-0.066	**0.071**	-0.068	-0.001	-0.199
10	-0.119	-0.099	**0.058**	-0.104	-0.007	-0.267
15	-0.215	-0.257	**0.401**	-0.164	-0.073	-0.287
20	-0.211	-0.366	**0.432**	-0.214	-0.167	-0.352
25	-0.320	-0.438	**0.369**	-0.247	-0.295	-0.283
30	-0.458	-0.497	**0.253**	-0.275	-0.335	-0.446
35	-0.470	-0.610	**0.181**	-0.306	-0.387	-0.480
40	-0.519	-0.666	**-0.189**	-0.322	-0.430	-0.542
45	-0.495	-0.742	**-0.296**	-0.357	-0.467	-0.737
50	-0.540	-0.808	-0.573	**-0.386**	-0.516	-0.727

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.135	-0.148	**0.007**	-0.128	-0.074	-0.278
10	-0.256	-0.230	**-0.103**	-0.225	-0.128	-0.403
15	-0.413	-0.446	**0.203**	-0.346	-0.237	-0.461
20	-0.473	-0.572	**0.205**	-0.431	-0.399	-0.595
25	-0.596	-0.671	**0.102**	-0.490	-0.572	-0.516
30	-0.716	-0.706	**-0.088**	-0.529	-0.650	-0.731
35	-0.711	-0.778	**-0.263**	-0.573	-0.702	-0.728
40	-0.731	-0.796	-0.678	**-0.597**	-0.703	-0.727
45	-0.701	-0.819	-0.681	**-0.635**	-0.710	-0.873
50	-0.730	-0.843	-0.780	**-0.672**	-0.726	-0.803

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 277,020 / 664,548 (41.7%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.675	0.653	**0.693**	0.667	0.664	0.668
10	0.673	0.651	**0.685**	0.664	0.668	0.668
15	0.670	0.651	**0.676**	0.663	0.667	0.667
20	0.669	0.651	**0.672**	0.662	0.664	0.666
25	0.667	0.652	**0.671**	0.662	0.663	0.665
30	0.665	0.652	**0.670**	0.661	0.663	0.665
35	0.664	0.652	**0.669**	0.660	0.663	0.664
40	0.664	0.652	**0.667**	0.660	0.663	0.664
45	0.663	0.652	**0.667**	0.659	0.663	0.663
50	0.662	0.652	**0.666**	0.659	0.663	0.663

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.980	**0.991**	0.987	0.982	0.983	0.986
10	0.981	**0.988**	0.983	0.981	0.984	0.983
15	0.982	**0.986**	0.985	0.980	0.984	0.983
20	0.982	**0.985**	0.985	0.981	0.983	0.983
25	0.983	**0.985**	0.984	0.981	0.982	0.983
30	0.983	**0.985**	0.983	0.981	0.982	0.983
35	0.984	**0.985**	0.983	0.981	0.982	0.982
40	0.984	**0.984**	0.983	0.981	0.983	0.982
45	0.983	**0.984**	0.984	0.981	0.983	0.982
50	0.983	0.984	**0.984**	0.982	0.982	0.982

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 277,020 / 664,548 (41.7%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.037	**0.244**	0.180	0.086	0.170	0.143
10	0.186	0.271	0.318	0.160	**0.325**	0.274
15	0.188	0.196	**0.655**	0.201	0.376	0.367
20	0.224	0.247	**0.677**	0.229	0.437	0.445
25	0.299	0.181	**0.618**	0.248	0.487	0.482
30	0.354	0.147	**0.506**	0.251	0.458	0.493
35	0.347	0.136	**0.547**	0.268	0.472	0.504
40	0.375	0.147	**0.537**	0.271	0.470	0.511
45	0.370	0.233	**0.540**	0.270	0.467	0.508
50	0.334	0.204	**0.503**	0.283	0.476	0.499

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.056	**0.168**	0.136	0.029	0.075	0.076
10	0.021	0.158	**0.201**	0.040	0.154	0.150
15	-0.043	0.044	**0.522**	0.020	0.135	0.190
20	-0.051	0.050	**0.560**	0.005	0.140	0.206
25	-0.020	-0.043	**0.509**	-0.006	0.154	0.194
30	-0.004	-0.093	**0.434**	-0.030	0.087	0.154
35	-0.041	-0.115	**0.440**	-0.039	0.068	0.122
40	-0.055	-0.101	**0.400**	-0.059	0.052	0.104
45	-0.096	-0.027	**0.361**	-0.078	0.030	0.088
50	-0.142	-0.052	**0.252**	-0.082	0.023	0.062

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 277,020 / 664,548 (41.7%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.150	**0.107**	0.076	-0.044	-0.010	-0.003
10	-0.159	0.042	**0.060**	-0.091	0.016	0.035
15	-0.300	-0.120	**0.367**	-0.148	-0.059	0.037
20	-0.376	-0.151	**0.379**	-0.210	-0.109	0.014
25	-0.351	-0.318	**0.288**	-0.253	-0.137	-0.034
30	-0.366	-0.491	**0.196**	-0.291	-0.234	-0.095
35	-0.416	-0.589	**0.079**	-0.311	-0.274	-0.133
40	-0.467	-0.670	**-0.126**	-0.334	-0.307	-0.155
45	-0.523	-0.700	-0.299	-0.368	-0.326	**-0.180**
50	-0.559	-0.768	-0.558	-0.368	-0.338	**-0.210**

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.226	**0.044**	0.013	-0.117	-0.092	-0.073
10	-0.284	**-0.081**	-0.101	-0.224	-0.139	-0.097
15	-0.465	-0.285	**0.172**	-0.341	-0.288	-0.153
20	-0.573	-0.386	**0.143**	-0.438	-0.386	-0.216
25	-0.551	-0.556	**0.010**	-0.496	-0.417	-0.291
30	-0.561	-0.748	**-0.171**	-0.551	-0.523	-0.375
35	-0.604	-0.804	**-0.290**	-0.578	-0.564	-0.421
40	-0.656	-0.815	-0.532	-0.613	-0.599	**-0.450**
45	-0.709	-0.805	-0.683	-0.646	-0.617	**-0.491**
50	-0.738	-0.831	-0.770	-0.660	-0.631	**-0.541**

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 140,118 / 664,548 (21.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.680	0.650	**0.693**	0.668	0.668	0.663
10	0.676	0.650	**0.685**	0.665	0.665	0.663
15	0.672	0.650	**0.676**	0.663	0.665	0.662
20	0.669	0.651	**0.672**	0.662	0.666	0.662
25	0.668	0.652	**0.671**	0.661	0.665	0.661
30	0.667	0.653	**0.670**	0.660	0.665	0.661
35	0.666	0.653	**0.669**	0.660	0.664	0.661
40	0.665	0.653	**0.667**	0.659	0.664	0.660
45	0.664	0.654	**0.667**	0.659	0.664	0.660
50	0.664	0.654	**0.666**	0.659	0.664	0.660

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.983	0.987	**0.987**	0.987	0.984	0.985
10	0.984	**0.986**	0.983	0.984	0.983	0.983
15	0.984	**0.985**	0.984	0.985	0.983	0.981
20	0.984	**0.985**	0.984	0.985	0.982	0.981
25	0.983	**0.985**	0.984	0.984	0.981	0.982
30	0.983	**0.984**	0.983	0.984	0.981	0.981
35	0.983	**0.984**	0.983	0.984	0.982	0.981
40	0.982	**0.984**	0.983	0.984	0.982	0.981
45	0.982	0.983	**0.983**	0.983	0.982	0.982
50	0.982	0.983	**0.984**	0.983	0.982	0.982

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 140,118 / 664,548 (21.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.136	0.162	**0.186**	0.114	0.150	0.056
10	0.296	0.160	0.306	0.183	**0.320**	0.116
15	0.382	0.138	**0.611**	0.235	0.338	0.160
20	0.347	0.086	**0.671**	0.272	0.373	0.172
25	0.381	0.089	**0.635**	0.280	0.362	0.190
30	0.412	0.036	**0.493**	0.304	0.381	0.209
35	0.429	0.049	**0.507**	0.324	0.446	0.219
40	0.431	0.097	0.415	0.333	**0.457**	0.220
45	0.414	0.124	0.378	0.330	**0.494**	0.241
50	0.448	0.122	**0.504**	0.336	0.504	0.252

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.076	0.093	**0.140**	0.048	0.082	-0.021
10	0.169	0.035	0.191	0.052	**0.193**	-0.032
15	0.198	-0.032	**0.484**	0.034	0.151	-0.054
20	0.105	-0.123	**0.574**	0.022	0.123	-0.080
25	0.076	-0.149	**0.523**	0.001	0.067	-0.091
30	0.052	-0.217	**0.417**	0.001	0.038	-0.100
35	0.046	-0.222	**0.419**	-0.001	0.066	-0.116
40	0.036	-0.196	**0.275**	-0.021	0.058	-0.138
45	0.001	-0.170	**0.233**	-0.035	0.067	-0.147
50	0.014	-0.190	**0.252**	-0.043	0.036	-0.159

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 140,118 / 664,548 (21.1%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.010	0.032	**0.071**	-0.019	0.018	-0.090
10	0.047	-0.079	0.050	-0.062	**0.078**	-0.162
15	0.026	-0.192	**0.351**	-0.098	-0.020	-0.241
20	-0.120	-0.361	**0.395**	-0.142	-0.075	-0.294
25	-0.171	-0.432	**0.339**	-0.197	-0.148	-0.347
30	-0.192	-0.519	**0.214**	-0.236	-0.202	-0.384
35	-0.185	-0.601	**0.138**	-0.265	-0.194	-0.423
40	**-0.193**	-0.669	-0.289	-0.303	-0.214	-0.446
45	**-0.229**	-0.688	-0.342	-0.336	-0.230	-0.461
50	**-0.215**	-0.743	-0.561	-0.358	-0.276	-0.473

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.065	-0.034	**0.007**	-0.088	-0.043	-0.159
10	-0.073	-0.222	-0.107	-0.192	**-0.048**	-0.296
15	-0.139	-0.382	**0.158**	-0.286	-0.189	-0.418
20	-0.319	-0.566	**0.158**	-0.364	-0.266	-0.516
25	-0.390	-0.674	**0.072**	-0.441	-0.336	-0.578
30	-0.456	-0.768	**-0.105**	-0.486	-0.381	-0.618
35	-0.501	-0.814	**-0.238**	-0.535	-0.390	-0.655
40	-0.553	-0.810	-0.650	-0.578	**-0.429**	-0.692
45	-0.610	-0.761	-0.638	-0.616	**-0.470**	-0.708
50	-0.613	-0.782	-0.716	-0.638	**-0.541**	-0.717

Seed 2024; single-run values. No across-seed standard deviation is available.